Simplifying algebraic expressions

1 For each of the following, identify the numerical coefficient of the given variable.

a $-18y$ b $-k$ c $\frac{t}{3}$ d $\frac{6m}{7}$

2 Consider the following expression.
Which is the like terms with $-3x^2$?

$$5x^3 - 3x^2 - 3x + 10 + 9x^2$$

3 Determine whether the following expressions is equivalent to $2(w+8)$?

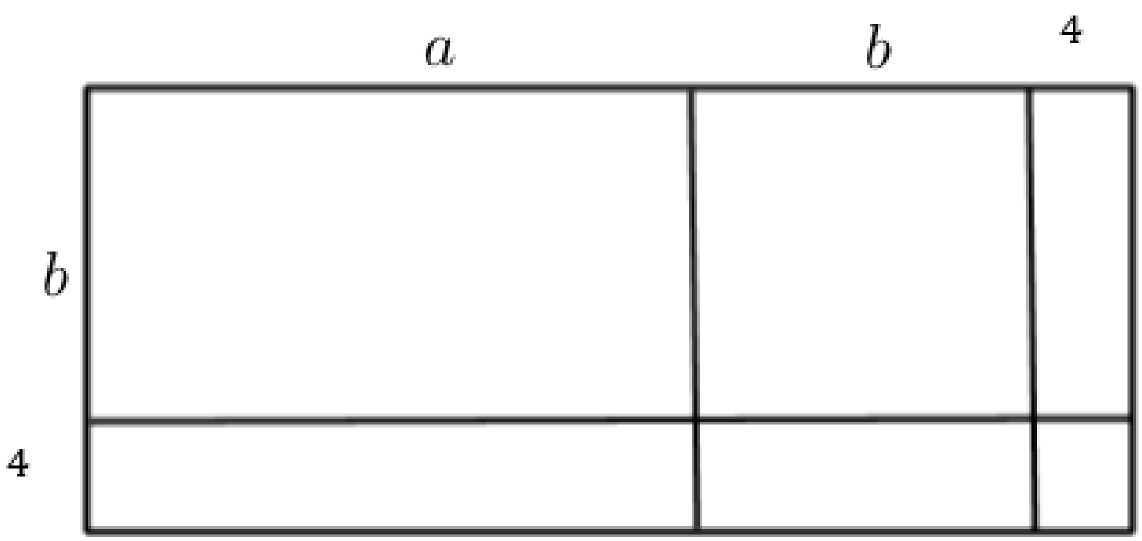
a $w + 8 \times 2$ b $(w+8)+(w+8)$

c $2w + 16$ d $w + w + 16$

4 Determine whether each of the following expressions are equivalent to $\frac{x+10}{5}$.

a $\frac{x}{5} + 2$ b $\frac{x}{5} + 10$ c $x + 2$ d $x + 10$

5 Determine whether each of the following expressions could be interpreted as representing the area of the given shape.



a $(a+b+4)(b+4)$ b $(a+b)b + 4^2$

c $(a+b)(b+4) + 4(b+4)$ d $4b(a+b+4)$

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6 Simplify the following expressions:

a $\dfrac{20n}{5}$

b $8a \times 3b$

c $8x - 9x - 10x$

d $-2m - 9m$

e $(-9n) \times 2$

f $7x + 7y - 3y - 6x$

g $8j - 6k + 5 - 6j + 6k - 3$

h $\dfrac{15yw}{5y}$

i $\dfrac{-14a}{2}$

j $(-24pq) \div 2p$

k $-10m^2 + 9m + 20m^2 - 3m^2$

l $(-3u) \times (-6a) \times (-4t)$.

m $13m - 4m \times 2$

n $(y - 9) + (y - 2)$

o $15x + (13x - 8x)$

p $9x - (7x - 3x)$

q $-3x - (-4 - 5x)$.

r $2y - 8(y - 5) - 7$

s $10x - 4x \times 4 + 9x$

t $4xy + 7z + 3 - (2 - 8yx - 9z)$

u $(3x + 5y + 8) - (7x - 9)$

v $9 - (p - 6)$

w $5p \times 3 - ((-6p) \div 3)$

7 Distribute and simplify the following expressions:

a $2(4x + 3x)$

b $5(y - 7) - 3y - 1$

c $2y + 5 + 3(y + 9)$

d $-10(x - 8) + 1$

e $5(y - 3) + 3y + 1$

f $(9y + 6) + (7y - 7)$

g $3y + 4(y - 1) + 5$

h $4y + 4 - 5(y - 9)$

i $4(x - 9) + 8$

j $9(8x - 2) + 8$

k $9y - 7 - 7(y + 8)$

l $3 + 7(x + 6) + 12$

m $7 + 9(2x + 5) + 8$

n $5(y + 9) - 7(y - 4)$

o $-10(a - 10) + 8$

p $3(5(x - 4) + 2)$

q $-8\left(\dfrac{x}{3} + 10\right) + 3$

r $-6(x - y) - (2x + 5y)$

s $-(2 - 4(x - 6y)) + 2(5x + y)$

8 What is the result when $-3x + 6$ is taken away from $x + 8$?

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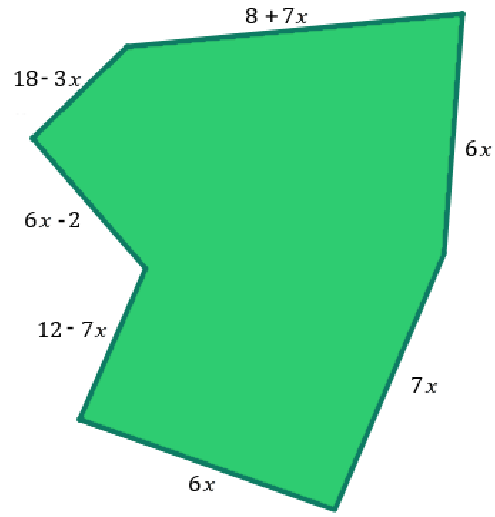
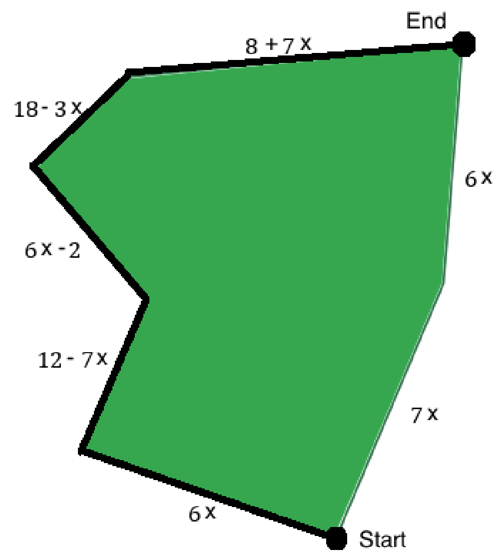
9 Kathleen's paddock has dimensions as shown in the diagram. All dimensions are in meters.

a Write a fully simplified expression for the perimeter of the paddock.

b On Mondays Kathleen runs 5 laps of the entire perimeter of the paddock. Write a fully simplified expression for the distance she runs each Monday.

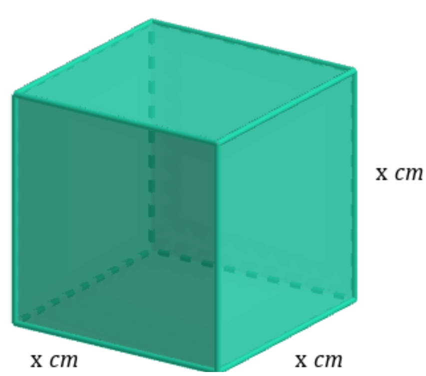


c On Saturdays Kathleen goes for a much shorter run. The route is shown in the diagram below by the thick, dark line. She runs from the Start to the End once. How much further does she run on Mondays than on Saturdays?



10 Express the area of a square with sides of 19 cm as a product of two numbers.

11 Express the surface area of a cube as a sum of the areas of each face.



12 In a particular year, Re-source Waste Recovery produces sewerage treatment plants at a cost of $\$u$ per plant. Re-source Waste Recovery also pays a fee of $\$b$ per year for its use of the production facilities.

a Using only addition, write an expression that represents the total production cost from 1 year if there are 3 sewerage treatment plants produced in that year.

b Write an expression equivalent to the production cost in that year.

13 A clothing store receives a shipment of jackets and skirts. The jackets come in blue boxes containing 6 jackets each, while the skirts come in red boxes containing 5 skirts each. In this shipment, there are x blue boxes and y red boxes.

a Write an expression representing the number of boxes in the shipment.

b Write an expression that represents the number of clothing items in the shipment.

14 Consider the case when n is even and m is odd. Each of the following expressions equates to a whole number, where $n > 2$. Determine whether each of the following is even.

a $3n + 2m$

b $8n + 3m + 3$

c $2n + 3m$

d $7n + 3m + 2$

e $m + n$

f $n + n + n + m + m$

15 Tom has to pay the price of his meal, $\$b$, plus a 10% tip.
Write an expression representing the price he will pay for his meal including the tip.

16 To simplify an expression, Christa writes down her working out below.

1 $9xy + x + 7 + x + y + xy + 3 = 9xy + x + x + y + xy + 10$

2 $= 9xy + 2x + y + xy + 10$

3 $= 9xy + 2xy + xy + 10$

4 $= 12xy + 10$

a At which step does Christa make a mistake?

b Christa asks for your help to simplify this expression. What should the answer actually be?

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17 To simplify an expression, Homer writes down his working out below.

$$\begin{aligned} \textcircled{1} \quad 2x^2 + 5y + 6x^2y + 3x^2y &= 2x^2y + 11x^2y + 3x^2y \\ \textcircled{2} \quad &= 13x^2y + 3x^2y \\ \textcircled{3} \quad &= 16x^2y \end{aligned}$$

a At which step does Homer first make a mistake?

b Homer asks for your help to simplify this expression. What should the answer actually be?

Additional questions

18 Simplify the following:

a $(5u - 2u) \times 3$

b $(13a + 15a) \div 7$

c $9p \div (6p - 3p)$

d $80mn \div (4m) \div (5n)$.

e $(-7n) \times 4 - (2n + 5n)$

f $(6u \times 12vw) \div (6v \times 32u)$

g $(19v - 5v) \div 2 \times 3$.

h $30r \div (3r + 2r) + 5r$.

i $5v - 5(6v - 2v)$

j $6v + (15v - (9v - 2v))$.

k $-8t - (8t - 6t \div 2)$

l $((-8r) \times 16st) \div (24s \times (-9r))$

m $(7m - 5m) \times 4m$

n $(-4n) \times 8 - (7n + 4n)$

o $\dfrac{16x}{3x + 5x}$

p $\dfrac{20s}{4s} \times 6t$

q $\dfrac{21m}{3} + \dfrac{10m}{2}$

r $9v \times 2 - \dfrac{25v}{5}$

s $15m + \dfrac{51m - m}{5m}$.

t $\dfrac{58r + 5r}{7r \times 3}$

u $\dfrac{7r + 5r}{7r - 5r}$

v $\dfrac{7rs + 23rs}{5r \times 3s}$

w $5n^2 + \dfrac{20n^2}{5}$

x $3m - \dfrac{12m^2}{3m}$.

y $5n \times 4n - \dfrac{8n^2}{4}$

z $\dfrac{7st + 53st}{4s \times 5t}$

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19 Simplify the following:

a $2t + 4 \times 8t$

b $2c + 7c \times 3 + 4c$

c $61a - 6 \times 9a - 2a$

d $8 \times 3a + 6 \times 2a$

e $3n^2 + 8n^2 - 3n^2$

f $9n^2 + 3n \times 6n$

g $5n^2 + 5n^2 \times 2 + 7n^2$

h $5mn \times 3n - 2mn^2$

i $20w - 10w \div 2 - 3w$

j $6p \times 8q \div 2$

k $60n^2 \div 5n \div 3$

l $8jk \times 15k \div 10j$

m $20pq \div 5q \times 2p$

n $12s - 8s \div 4$

o $12u \div 3 - 2u$

p $8jk \times 15k \div 20j$

q $\dfrac{8x}{12} + \dfrac{10x}{12}$

r $\dfrac{y}{11} - \dfrac{9y}{11}$

s $\dfrac{17y}{20} + \dfrac{14y}{20}$

t $\dfrac{7x}{3} - \dfrac{5x}{3}$

u $\dfrac{3x}{4} + \dfrac{3x+4}{4}$

v $\dfrac{p}{2} - \dfrac{p}{4}$

w $\dfrac{3p}{7} + \dfrac{2p}{35}$

x $\dfrac{p}{3} + \dfrac{6p}{15}$

y $\dfrac{x}{3} + \dfrac{x+2}{12}$

z $\dfrac{4x+3}{3} - \dfrac{x-3}{6}$

20 Consider $\dfrac{w}{3} + \dfrac{w}{9}$.

a Find the lowest common denominator of $\dfrac{w}{3}$ and $\dfrac{w}{9}$.

b Hence, write $\dfrac{w}{3} + \dfrac{w}{9}$ as a single fraction.

21 Consider $\dfrac{m}{4} - \dfrac{m}{12}$.

a Find the lowest common denominator of $\dfrac{m}{4}$ and $\dfrac{m}{12}$.

b Hence, write $\dfrac{m}{4} - \dfrac{m}{12}$ as a single fraction in simplest form.

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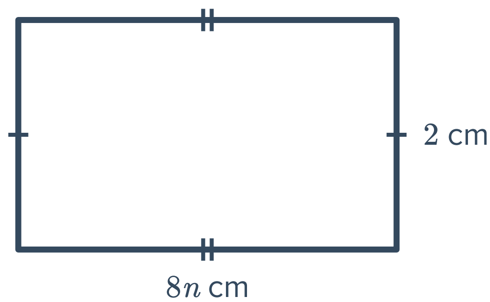
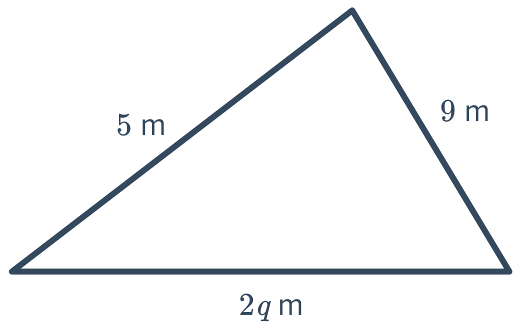
22 Write an algebraic expression for the perimeter of the following shapes:

a



b



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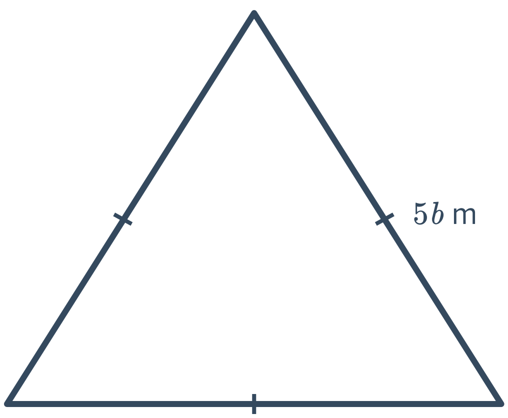

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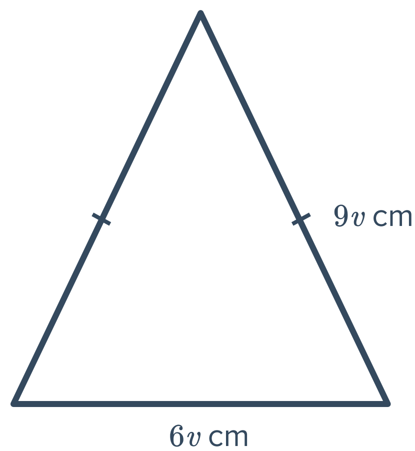

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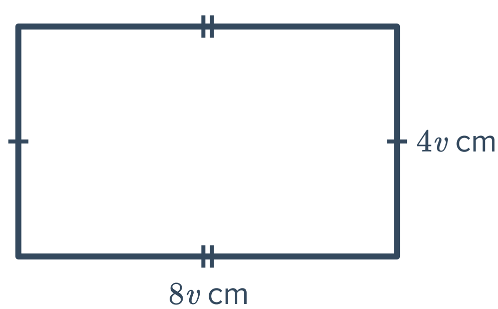

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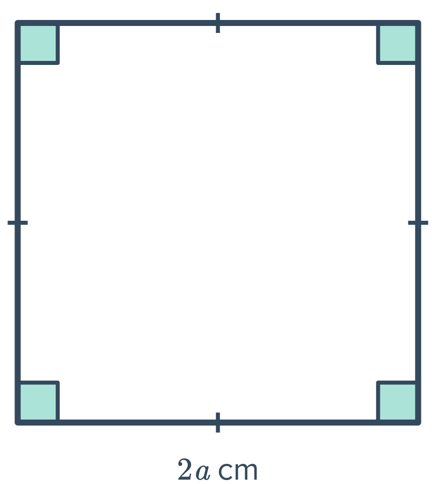

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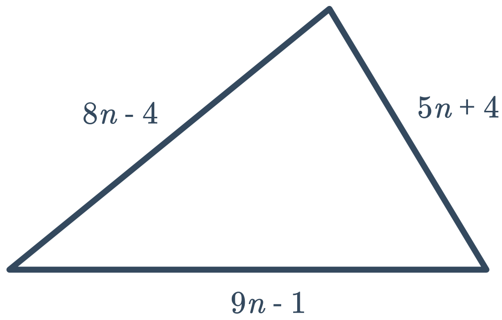
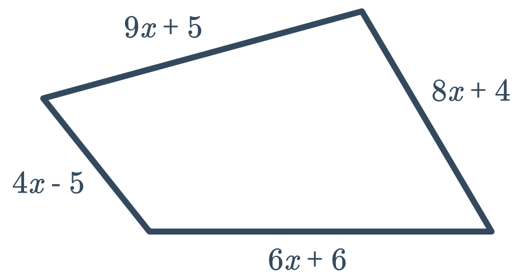
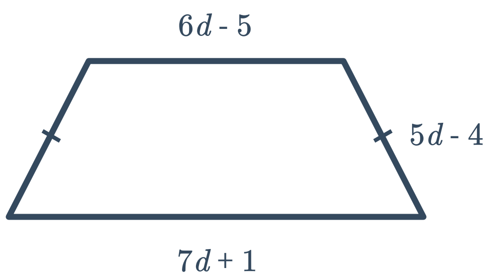
g



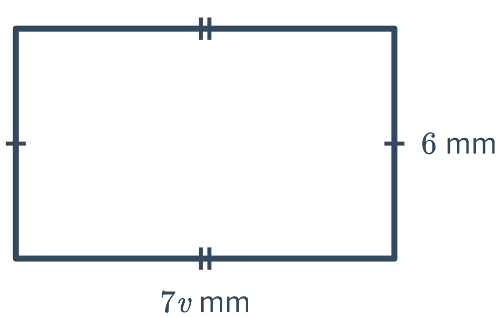


i

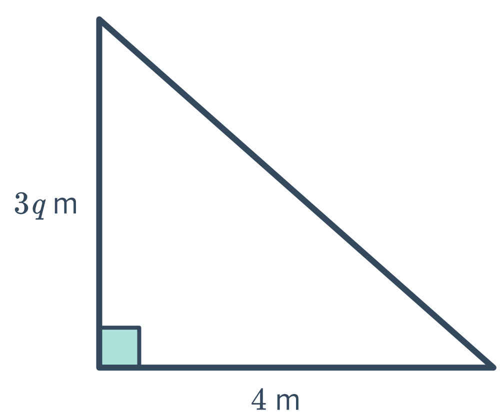


23 Write an algebraic expression for the area of the following shapes:

a



b



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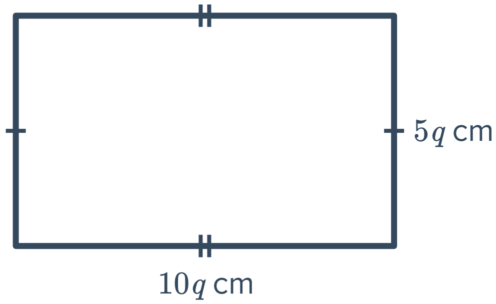
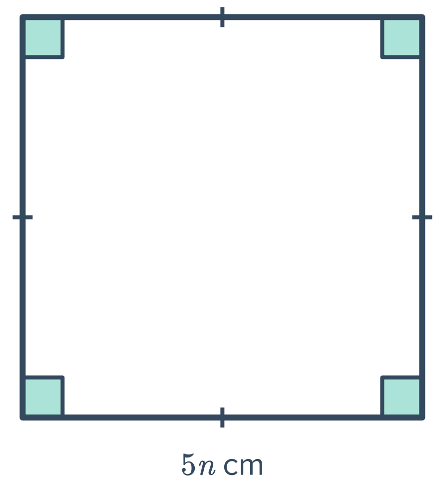
c





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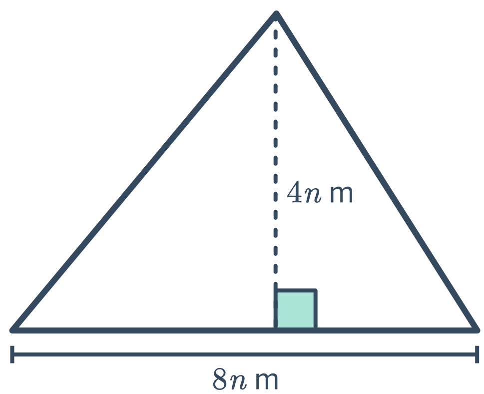
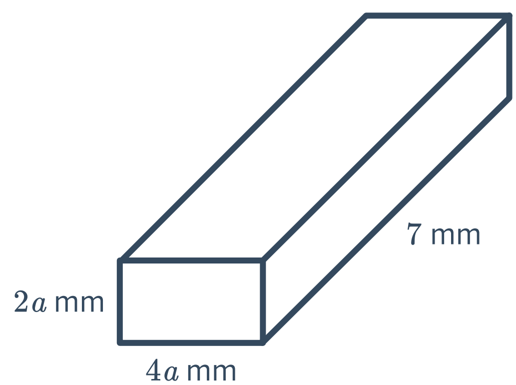


24 Consider the following rectangular prism:

a Write an expression for its volume.

b Write an expression for its surface area.



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25 Dylan, Jimmy and Valentina are travelling from Adelaide to Sydney at x, y and z $\mathrm{km/hr}$ respectively.
Let S_1 represent the average speed of Dylan and Jimmy's vehicles, and S_2 represent the average speed of Dylan and Valentina's vehicles.
Write a simplified expression for $S_1 + S_2$ in terms of x, y and z.

26 Frank wants to determine the profit he makes on each eraser he sells. Frank can sell 3 erasers for x dollars, and it costs y dollars to make 9 erasers.

a Determine a simplified expression for the profit Frank makes on each eraser. Leave your answer as a single fraction.

b Frank realised he could sell 3 erasers for $\$3$ more than they previously were, and reduce the cost of production of 9 erasers by $\$2$. Find the new profit per eraser.

27 Glen lived in a rectangular bedroom. His mother asked him to find out the area of his room, and since Glen did not have a measuring tape or ruler, he counted how many of his steps each wall was.
If Glen's foot length is x cm, and two adjacent walls were 12 and 10 footsteps long, write an expression for the area of his room.

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